

Zahara ERP Portal

Zero-Error cPanel Shared Hosting Deployment Guide

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Platform: cPanel Shared Hosting (LiteSpeed Web Server)

Database: SQLite 3 (Configured for Zero Data Loss)

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1. Architecture Design & Safety Principles

To ensure high security, seamless updates, and zero database loss on shared hosting, always implement the following layout rules:

- **Keep Code Private:** Clone repository files outside the public HTML folder (e.g., /home/username/zahara_erp/). Python code and secrets should never be in a directory browsable by url.
- **Isolate Database:** Create a dedicated folder (e.g., /home/username/zahara_erp_db/) for your SQLite database. This ensures pulling code updates from GitHub never overwrites or resets your database.
- **Symlink Assets:** The subdomain's root directory (/home/username/erp.zaharaflowers.com/) must contain only web server routing files (.htaccess) and symlinks pointing to static and media files inside your application directory.

2. Zero-Error Deployment Checklist

Phase A: Local Code Adjustments

1. Check **settings.py** to load the database path from environment variables, avoiding overwritten configurations:

```
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.sqlite3',
        'NAME': config('DATABASE_PATH', default=str(BASE_DIR / 'db.sqlite3')),
    }
}
```

2. Enable SSL and CSRF trusted origins to prevent form submit crashes in production:

```
ALLOWED_HOSTS = config('ALLOWED_HOSTS', default='localhost,127.0.0.1', cast=Csv())
CSRF_TRUSTED_ORIGINS = config('CSRF_TRUSTED_ORIGINS', default='http://localhost:8000', cast=Csv())
```

3. Verify that your **.gitignore** does not ignore required Django app folders (e.g. data_import). Ensure you only ignore spreadsheet subfolders:

```
backend/data_import/extracted_sheets/
backend/data_import/formatted_sheets/
```

Phase B: cPanel Domain and SSL Settings

1. **Create Subdomain:** Set up `erp.zaharaflowers.com` in cPanel. Note its exact document root folder (e.g. `/home/zaharaf1/erp.zaharaflowers.com`).
2. **Install SSL:** Go to cPanel -> SSL/TLS Status and run Let's Encrypt auto-SSL on the subdomain to secure forms and authentication.

Phase C: Directory Structure & File Manager Upload

1. In File Manager, create your database directory outside the codebase: `/home/zaharaf1/zahara_erp_db/`.
2. Upload your local database file `db.sqlite3` into this newly created folder.

Phase D: Git Cloning (Private Repository)

cPanel Git tools block URLs containing passwords/tokens. Use the following bypass format:

1. Create a **GitHub Personal Access Token (PAT)** classic with 'repo' scope.
2. Under cPanel -> **Git Version Control** -> click **Create**.
3. Paste the following URL, replacing your actual token (without username prefix):

```
https://ghp_YOUR_PERSONAL_ACCESS_TOKEN@github.com/RennieNjuguna/zahara_erp.git
```

4. Set directory to `/home/zaharaf1/zahara_erp` and click Create.
5. **CRITICAL LITESPEED STEP:** Right-click the newly cloned folder `/home/zaharaf1/zahara_erp` and set permissions to **755**. If set to `0700`, LiteSpeed will fail to read files and disable your Python application.

Phase E: Initialize the Python Application

1. Go to cPanel -> **Setup Python App** and click **Create Application**.
2. Configure the parameters exactly as follows:
 - **Python Version:** 3.10.x
 - **Application Root:** `zahara_erp/backend` (points to the subfolder containing `manage.py`)
 - **Application URL:** Select `erp.zaharaflowers.com`
 - **Application Startup File:** `passenger_wsgi.py`
 - **Application Entry Point:** `application`
3. Click **Create**.

Phase F: Configure Environment Variables

1. In File Manager, go to `/home/zaharaf1/zahara_erp/backend/`.
2. Create a file named `.env` and paste the production settings:

```
SECRET_KEY=your_production_secret_key
DEBUG=False
ALLOWED_HOSTS=erp.zaharaflowers.com
CSRF_TRUSTED_ORIGINS=https://erp.zaharaflowers.com
DATABASE_PATH=/home/zaharaf1/zahara_erp_db/db.sqlite3

# Email SMTP Settings
EMAIL_HOST=mail.zaharaflowers.com
EMAIL_PORT=465
EMAIL_USER=lucy@zaharaflowers.com
EMAIL_PASSWORD=your_email_password
EMAIL_IMAP_PORT=993
```

3. In Setup Python App, add requirements.txt to **Configuration files** and click **Run Pip Install**.

Phase G: Configure LiteSpeed Rewrite Rules

LiteSpeed ignores Passenger directives unless forced via rewrites. Update the subdomain `.htaccess`:

1. Open `/home/zaharaf1/erp.zaharaflowers.com/.htaccess` in File Manager.
2. Replace the contents to bypass static links and route all other traffic:

```
PassengerEnabled On
PassengerAppRoot "/home/zaharaf1/zahara_erp/backend"
PassengerBaseURI "/"
PassengerPython "/home/zaharaf1/virtualenv/zahara_erp/backend/3.10/bin/python"

RewriteEngine On
RewriteCond %{REQUEST_URI} !^/static/
RewriteCond %{REQUEST_URI} !^/media/
RewriteCond %{REQUEST_FILENAME} !-f
RewriteRule ^(.*)$ passenger_wsgi.py [QSA,L]
```

Phase H: Compile Static Files & Symlinks

1. Open `/home/zaharaf1/zahara_erp/backend/passenger_wsgi.py` in File Manager and temporarily set content to:

```
from deploy_helper import application
```

2. Restart the app in cPanel and load `https://erp.zaharaflowers.com` in your browser. Wait for the green success ticks.
3. Edit `passenger_wsgi.py` back to the default Django loading code and restart the Python app. Your system is now fully live!